

Automated and Explainable Outcome Prediction in NSCLC: Beyond Manual Feature Extraction using LLM Embeddings

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INTRODUCTION

- **Clinical notes** contain rich, unstructured **prognostic information**.
- **Manual extraction** is labor-intensive, slow, and **not scalable** for large cohorts.
- **LLM embeddings** can automate this process by generating context-aware features.
- **Explainability is essential** to translate AI predictions into **trustworthy clinical insights**.

GOALS

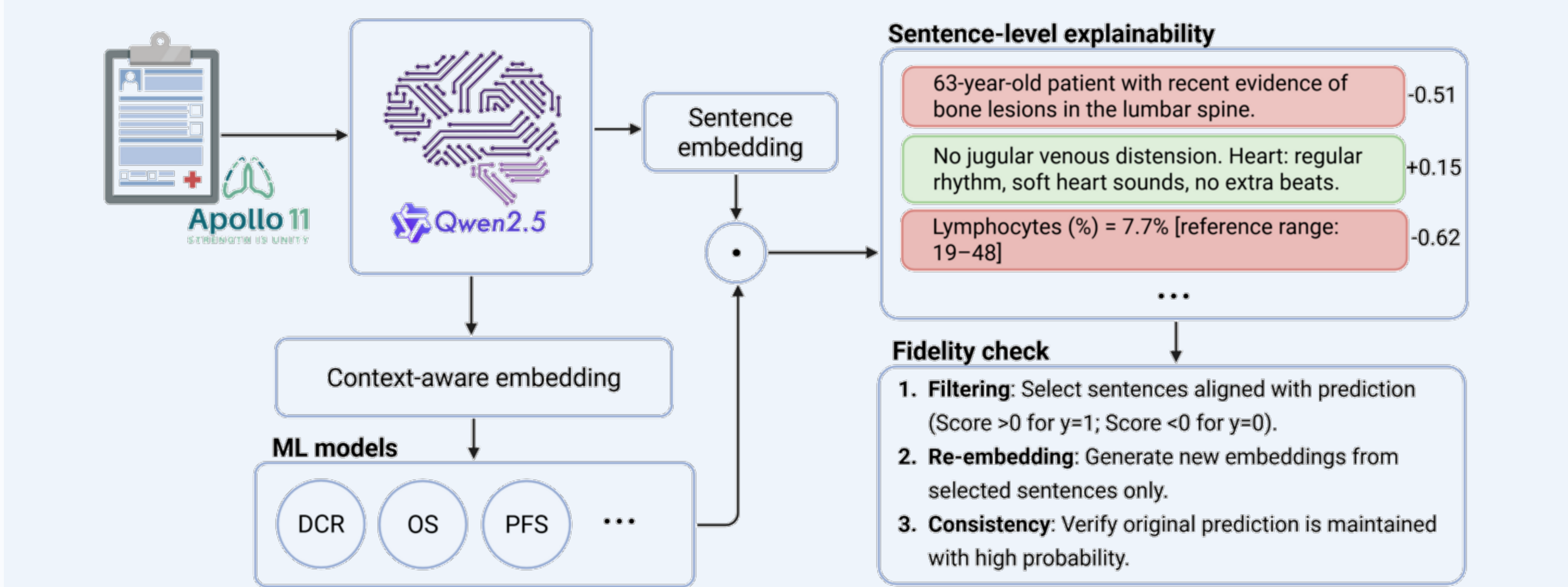
- **Automate** feature engineering from unstructured notes.
- **Predict** OS, PFS, DCR, and toxicity across **IO, CHT, and IO+CHT**.
- Provide **sentence-level explainability** for clinical transparency.
- **Validate** AI logic via a rigorous **Fidelity framework**.

DATASET

- **691 NSCLC patients**
 - INT Milan
- **Apollo 11** project
 - NCT05550961
- **Heterogeneous cohort** including patients treated with
 - **Immunotherapy**
 - **Chemotherapy**
 - **Immuno-Chemotherapy**
- Multimodal analysis of
 - unstructured **clinical reports**
 - **laboratory results**
 - **genomic profiling**

METHODS

- **Qwen2.5-7B** generates patient-level embeddings
- Implemented on a proprietary **on-premise server** to ensure data privacy and **GDPR compliance**
- Embeddings → prediction models
- Sentence-level explainability + fidelity



RESULTS

- Accurate prediction of **OS, PFS, DCR, toxicity**
- Explainability → key **clinical drivers**
- **Fidelity** validated explanations
- Fully **automated pipeline**

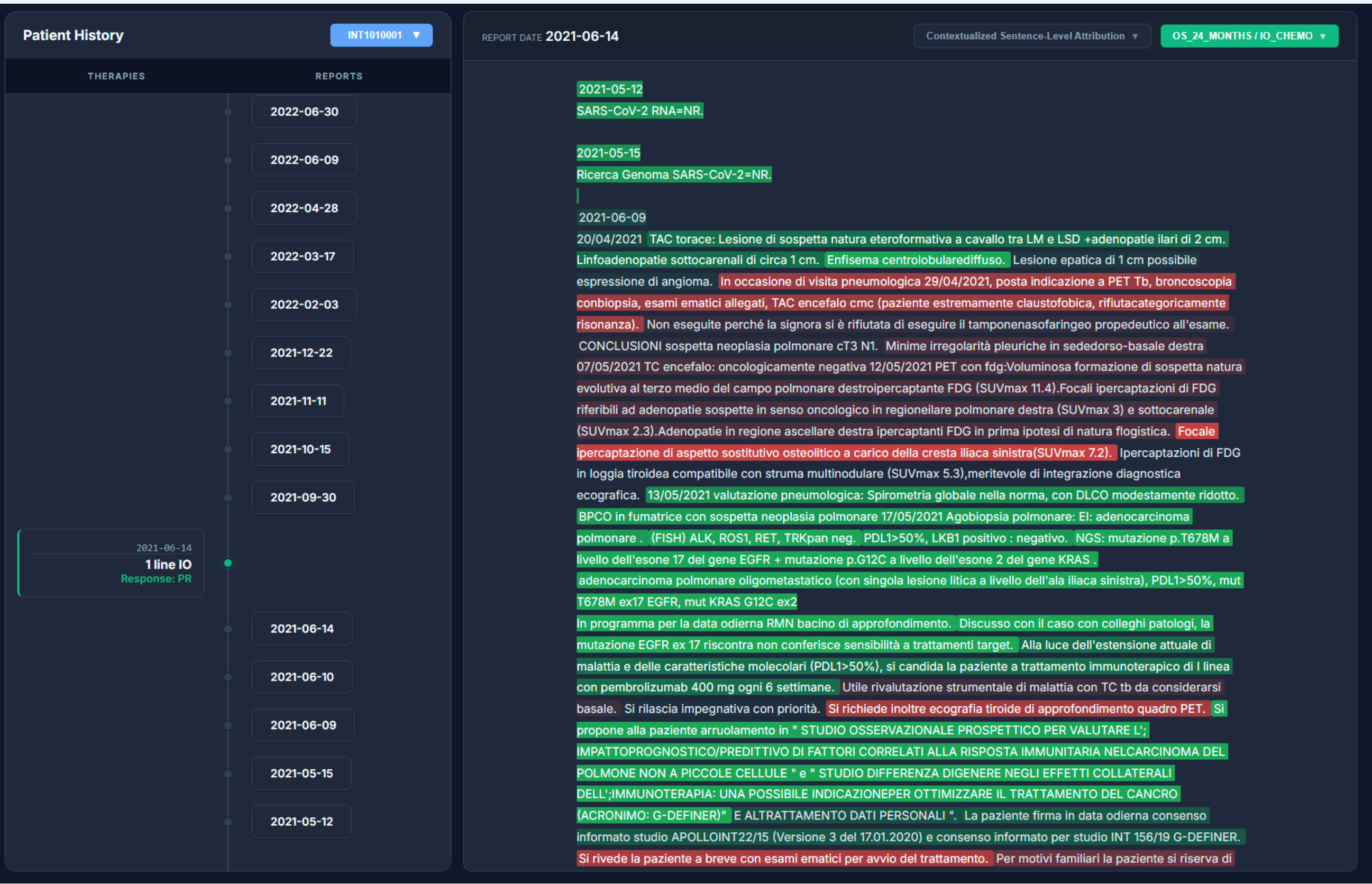
Outcome	IO	CHT	IO+CHT
OS 6	0.63 ± 0.13	0.79 ± 0.19	0.72 ± 0.20
OS 24	0.67 ± 0.16	0.81 ± 0.19	0.62 ± 0.22
DCR	0.55 ± 0.14	0.73 ± 0.19	0.56 ± 0.21
TOXICITY	0.57 ± 0.15	*	0.57 ± 0.24

Table 1: 5-fold CV AUC (Mean ± CI)

Outcome	IO	CHT	IO+CHT
OS	0.58 ± 0.04	0.68 ± 0.06	0.62 ± 0.06
PFS	0.61 ± 0.08	*	0.85 ± 0.10

Table 2: 5-fold CV C-index (Mean ± CI)

* Not assessed due to insufficient sample size for robust training.



CONCLUSION

- LLM embeddings replace manual feature engineering
- Transparent and reliable predictions
- Scalable for real-world clinical use